Interpretable Dominant-Taxon States for Characterizing IBD and Related Gut Microbiome Dysbiosis

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Abstract

Inflammatory bowel disease (IBD) has repeatedly been associated with heterogeneous gut microbiome dysbiosis, including loss or redistribution of obligate anaerobic commensals and expansion of facultative or inflammation-associated taxa. The practical challenge addressed here is not whether these broad taxonomic shifts occur, but whether they can be summarized as sample-level ecological states that remain interpretable across large cohorts. Gut enterotypes provide one such summary, but they are clustering-defined constructs whose labels depend on choices about data transformation, distance metric, clustering model, and the number of clusters. We therefore applied dominant-community state types (dCSTs), a deterministic rank-based description of retained-feature dominance, to 30,290 quality-controlled American Gut Project stool samples from a 34,711-sample SILVA gut abundance table. Self-reported IBD showed the strongest phenotype-level association with the dominance-state hierarchy. The central IBD result was that the familiar broad *Bacteroides*-associated dysbiosis pattern resolved into an ordered *Bacteroides*/Lachnospiraceae/*Alistipes* dominance-lineage, providing a more specific sample-level description of IBD-associated ecological context rather than implying that the named taxa are newly discovered IBD markers. Bayesian models also provided stabilized estimates for sparse *Morganella* and *Proteus* states, which are treated as exploratory enterobacterial signals. The IBD dominance-state summary was robust to a stricter control definition and to coarse antibiotic-status sensitivity analyses. External validation separated two forms of reproducibility: sample-level Halfvarson 2017 analyses supported both rebuilt-cohort and direct AGP-derived label-transfer signals, PRJEB84421 provided narrower rebuilt inflammatory support, and pooled HMP2 retained a weaker depth-2 transferred IBD signal that persisted in a first-sample-per-subject sensitivity analysis. IBS and autoimmune disease showed weaker but directionally related dominance patterns. These results support dCSTs as an interpretable framework for organizing established IBD-associated dysbiosis into deterministic dominance-lineage states, while broader gut-outcome applications and universal label portability remain exploratory.

Keywords: gut microbiome; inflammatory bowel disease; dysbiosis; enterotypes; dominant-community state types; Bayesian modeling

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Introduction

For common gut disorders, clinical phenotypes do not map neatly onto one standalone molecular or laboratory test. IBD diagnosis integrates clinical presentation with biochemical, stool, endoscopic, imaging, and histologic evidence rather than relying on one reference standard (Maaser et al. 2019). IBS is even more symptom-defined, with no validated diagnostic biomarker and a guideline-supported positive diagnostic strategy (Lacy et al. 2021). This creates a substantial interpretive challenge for microbiome studies: disease-associated community shifts are repeatedly observed, but they are heterogeneous and difficult to translate into sample-level ecological labels that can be compared across cohorts.

In IBD, the existence of microbiome dysbiosis is no longer the central question. Landmark studies have shown that IBD is not characterized by one universal microbial signature, but often involves reduced diversity, depletion or redistribution of butyrate-associated obligate anaerobes, and expansion of facultative or inflammation-associated taxa (Frank et al. 2007; Gevers et al. 2014; Lloyd-Price et al. 2019; Sokol et al. 2008; Manichanh et al. 2006). These observations motivate a practical representation question rather than a new diagnostic-test claim: can established but heterogeneous IBD-associated microbiome shifts be summarized into transparent sample-level dominance states that can be compared across cohorts?

Gut enterotypes were introduced to provide such labels by compressing high-dimensional gut abundance profiles into a small number of clustering-defined configurations (Arumugam et al. 2011). That abstraction can be informative for broad community structure, but it has remained controversial precisely because the labels are clustering constructs: assignments depend on data transformation, distance metric, initialization, model class, and the chosen number of clusters (Koren et al. 2013; Knights et al. 2014; Costea et al. 2018). Those choices can make it hard to determine what a label means biologically for a particular sample or whether labels from different analyses are directly comparable.

We use dominant-community state types (dCSTs) as a deterministic rank-based framework for naming sample-level gut ecological states. A dCST is defined by the within-sample ordering of retained taxa. At depth 1, the state is assigned by the dominant retained taxon; at depth 2, it is refined by the ordered pair of the two most abundant retained taxa. Because assignment depends only on within-sample rank order, the same label is recovered from counts, proportions, or any sample-wide rescaling that preserves taxon order within a sample. We use *dominance-lineage* to denote the ordered path defined by a depth-ℓ dCST label as the hierarchy is refined across ranks. The term refers only to within-sample dominance order and does not imply phylogenetic relatedness. The dCST construction is therefore not intended to replace mechanistic microbiome modeling. Instead, it provides a transparent way to ask whether IBD-associated dysbiosis can be summarized as dominance states that remain interpretable after covariate adjustment, hierarchical refinement, Bayesian sparse-state follow-up, and external validation. The construction is illustrated in Figure 1.

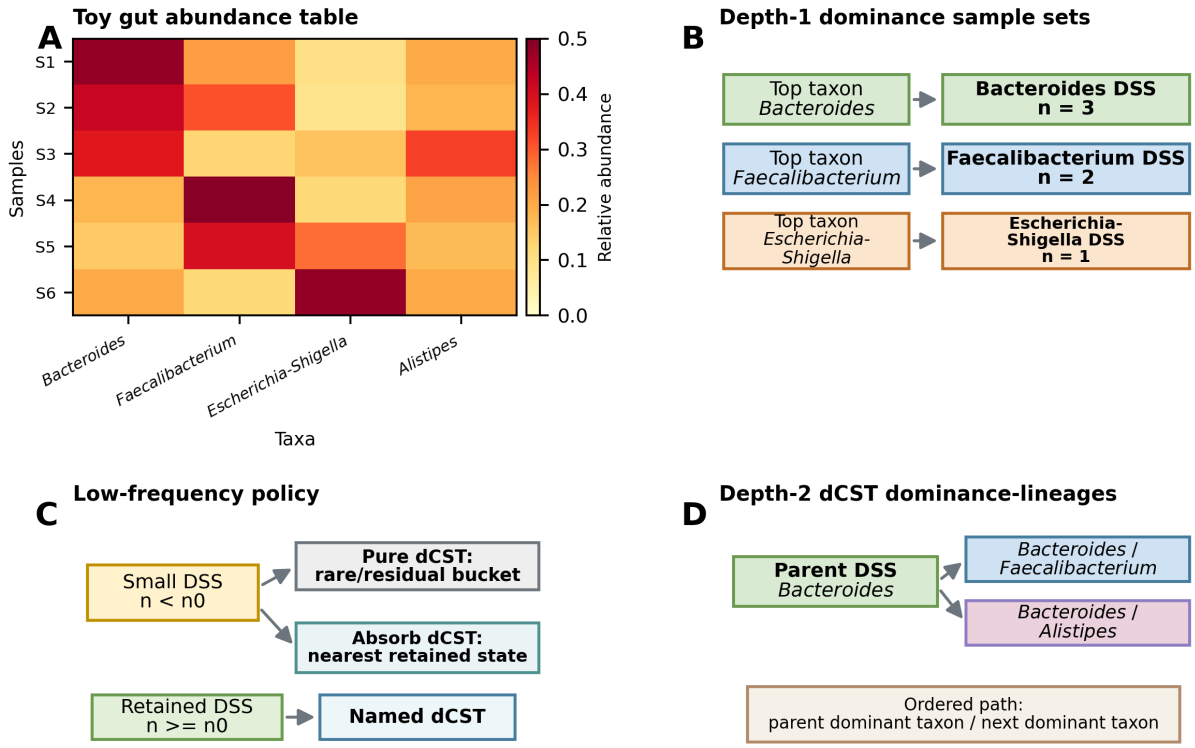

Figure 1: Conceptual schematic of dCST construction. Panel A shows a toy gut abundance table, and panel B groups samples into dominance sample sets (DSSs) according to their top-ranked retained taxon. Panel C contrasts the pure-policy view, in which low-frequency DSSs collapse into a residual bucket, with the absorb-policy view used here, in which low-frequency patterns are merged into retained named states. Panel D shows that a depth-2 label is an ordered dCST dominance-lineage, such as *Bacteroides*/*Faecalibacterium*, rather than a fitted cluster or a phylogenetic relationship.

This application therefore centers on IBD, with IBS and autoimmune disease included as related gut or immune-associated outcomes because literature suggests that shifts in dominant taxa or dominant taxon pairs can mark altered ecological states in those settings (Tap et al. 2017; Pittayanon et al. 2019; Scher et al. 2013; Vatanen et al. 2018). We use "inflammatory" as a disease-contextual and literature-grounded descriptor rather than as evidence that host inflammatory biomarkers were measured in the AGP discovery cohort. In this study we use the absorb low-frequency policy, which merges very small dominance patterns into larger named states rather than pooling them into one catch-all rare category. Pure-policy dCST results are used only as a baseline comparison in the supplement rather than as a second main-text results analysis.

Figure 2 previews how the dCST framework organizes the IBD-centered result: self-reported IBD had the strongest depth-resolved association with the dominance-state hierarchy, with a broad *Bacteroides* state resolving into more specific *Bacteroides*-centered dominance-lineages and the strongest external support coming from IBD-focused validation cohorts.

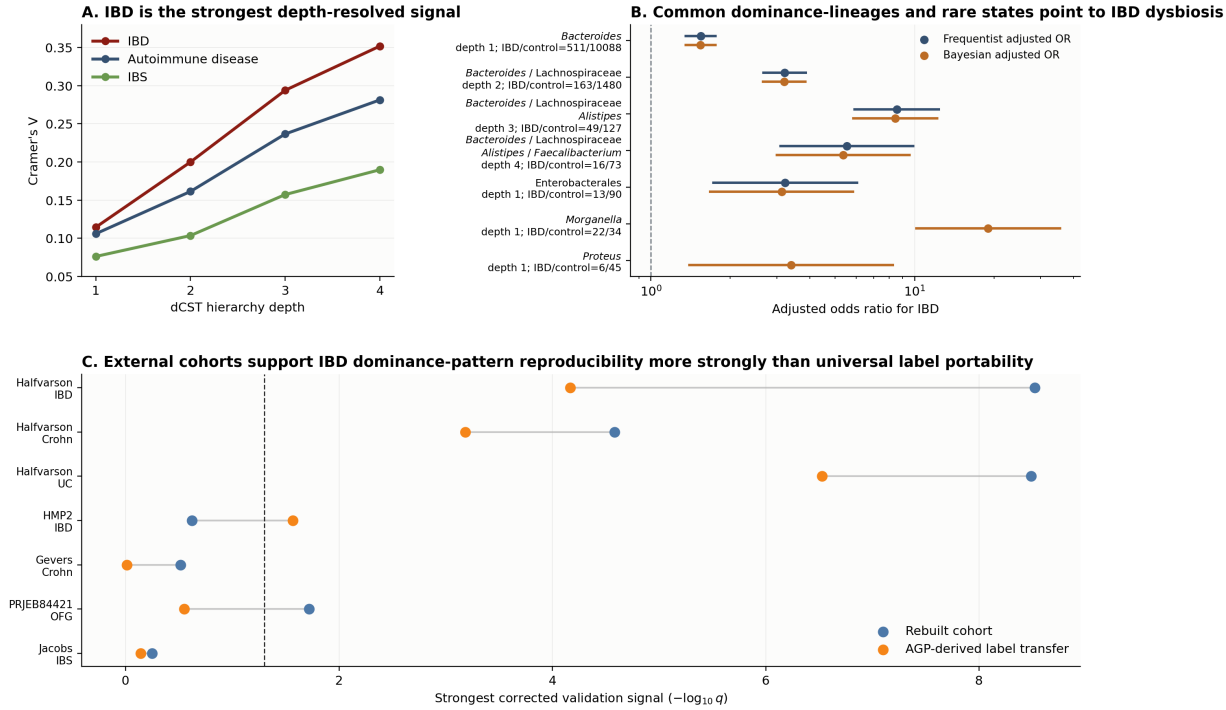

Figure 2: Overview of IBD-associated dominance-state structure. Panel A shows that IBD had the strongest depth-resolved absorb dCST association among the main gut or immune-associated phenotype analyses. Panel B summarizes representative common IBD dominance-lineages and sparse enterobacterial states using frequentist and Bayesian adjusted odds ratios. Panel C summarizes sample-level external validation: rebuilt-cohort analyses test whether disease-associated dominance structure is rediscovered within each cohort, whereas direct transfer of AGP-derived absorb dCST labels tests whether discovery-cohort labels remain informative without rebuilding the hierarchy. Subject-level validation sensitivity is reported in the text and supplement. The dashed line marks q = 0.05.

Results

Outcome-wide discovery overview

The discovery analysis asked whether interpretable dominance states could organize disease-associated gut ecology in 30,290 quality-controlled AGP stool samples. The absorb dCST hierarchy contained 40 named states at depth 1, 153 at depth 2, 479 at depth 3, and 744 at depth 4. The largest depth-1 states were *Bacteroides* (10,599 samples), *Escherichia-Shigella* (6,913), *Prevotella_9* (1,926), Enterobacteriaceae (1,596), and *Faecalibacterium* (1,405). The twelve most common depth-1 absorb dCST dominance-lineages covered 87.3% of the cohort (Supplementary Table S5), showing that a limited number of dominant ecological configurations accounted for most samples even after low-frequency patterns were absorbed into larger named states. Some retained-feature labels also carried low-resolution or potential non-gut annotations, including *Unassigned*, *Pseudomonas*, *Staphylococcus*, and, at deeper levels, *Eukaryota*/*Unassigned*/Bacteria/Mitochondria (Supplementary Tables S2 and S5). We retained these labels in the unsupervised hierarchy for transparent accounting, but they are not used as primary biological evidence for the IBD interpretation.

The depthwise phenotype screen showed that IBD provided the clearest outcome-level association with the dominance-state hierarchy. Across depths 1 to 4, Cramer's V increased from 0.11 to 0.35 in IBD, compared with 0.11 to 0.28 in autoimmune disease and 0.08 to 0.19 in IBS. Because IBD combined the largest depthwise growth in effect size, the clearest dominance-lineage-level interpretability, and the strongest external validation, we focus the detailed follow-up on IBD. IBS and autoimmune disease are retained as related secondary analyses, while additional non-IBD examples are summarized in the supplement (Supplementary Table S6). The absorb low-frequency policy most affected deeper levels of the hierarchy and is summarized separately in Supplementary Table S2.

IBD-associated dominance-lineage structure

Self-reported IBD provides the most clinically grounded inflammatory phenotype in the AGP screen, even though the discovery metadata do not separate treatment-naive Crohn disease, ulcerative colitis, disease activity, and medication exposure.

At the label level, the primary adjusted depth-1 screen showed enrichment of *Bacteroides* (adjusted OR 1.54, q = 8.1e-10), *Bifidobacterium* (adjusted OR 3.38, q = 2.2e-04), Enterobacterales (adjusted OR 3.23, q = 0.0016), and Lachnospiraceae (adjusted OR 1.99, q = 0.0030), together with depletion of *Prevotella_9* (adjusted OR 0.44, q = 2.2e-04). The Bayesian layer recovered the same common-state picture, but it also highlighted very rare enterobacterial states that were too small for the primary adjusted frequentist screen. In particular, *Morganella* showed a Bayesian adjusted OR of 18.99 (95% credible interval 10.15–35.53) and *Proteus* showed a Bayesian adjusted OR of 3.40 (95% credible interval 1.40–8.27), both with near-complete posterior support for enrichment in IBD. These rare-state examples were small: *Morganella* contained 22 IBD samples and 34 controls, and *Proteus* contained 6 IBD samples and 45 controls. We therefore treat them as exploratory facultative-anaerobe or Enterobacterales-linked rare-state signals rather than as standalone biomarkers or as evidence of disease severity.

Hierarchical refinement organized the broad IBD-associated pattern into more specific dominance-lineage summaries. At depth 2, *Bacteroides*/Lachnospiraceae reached adjusted OR 3.21 (q = 2.6e-33), *Bacteroides*/[*Eubacterium*] *eligens* group reached adjusted OR 22.50 (q = 5.0e-24), and *Bacteroides*/Enterobacteriaceae reached adjusted OR 2.73 (q = 2.2e-04). At depth 3, the strongest dominance-lineage was *Bacteroides*/Lachnospiraceae/*Alistipes* (adjusted OR 8.56, q = 1.5e-28), followed by *Bacteroides*/*[Eubacterium]_eligens_group*/Lachnospiraceae (adjusted OR 22.50, q = 5.0e-24) and *Bacteroides*/*Alistipes*/Lachnospiraceae (adjusted OR 5.95, q = 6.1e-15). At depth 4, the best supported dominance-lineage remained *Bacteroides*/Lachnospiraceae/*Alistipes*/*Faecalibacterium* (adjusted OR 5.53, q = 1.1e-07). Together, these deeper states indicate that the broad depth-1 *Bacteroides* state contains multiple IBD-associated dominance-lineages. The absorb policy preserves these dominance-lineages as named states rather than collapsing them into a generic rare-state category. A dominance-lineage label records rank context rather than a claim that every named component is enriched in IBD. For example, the depth-4 *Bacteroides*/Lachnospiraceae/*Alistipes*/*Faecalibacterium* state remains a *Bacteroides*-first retained-feature

context; *Faecalibacterium* is a lower-ranked component within that label (Supplementary Table S8). This distinction is important because the broader IBD literature often reports depletion of *F. prausnitzii* or butyrate-associated commensals.

The IBD-associated dominance-state summary was also robust to a stricter control definition. After removing non-IBD controls who carried any of the other AGP phenotype strings analyzed in this study (IBS, autoimmune disease, acid reflux, or seasonal allergies), the analysis retained all 1,166 IBD cases and 16,175 cleaner controls. The broad *Bacteroides* signal remained enriched (adjusted OR 1.59, q = 1.1e-10), as did the *Bacteroides*/Lachnospiraceae dominance-lineage at depth 2 (adjusted OR 3.15, q = 1.5e-28) and the *Bacteroides*/Lachnospiraceae/*Alistipes* dominance-lineage at depth 3 (adjusted OR 10.15, q = 6.2e-22). The depth-4 *Bacteroides*/Lachnospiraceae/*Alistipes*/*Faecalibacterium* dominance-lineage retained a finite Bayesian estimate but became too small for the primary adjusted frequentist screen. The rare *Morganella* and *Proteus* states likewise fell below the prespecified support threshold in this sensitivity analysis, reinforcing the need to interpret them cautiously (Supplementary Table S7).

Coarse antibiotic-status sensitivity analyses supported the same calibration. Among age-, sex-, and BMI-complete samples, adding the AGP `Antibiotic_Status` field as a covariate or excluding samples reporting antibiotic use in the previous week, month, or six months did not attenuate the leading *Bacteroides*-centered IBD signals. The depth-2 *Bacteroides*/Lachnospiraceae dominance-lineage, for example, had adjusted OR 3.21 in the age/sex/BMI model, 3.26 after including coarse antibiotic status, and 3.44 after excluding recent antibiotic users. These analyses do not replace detailed medication, treatment, or disease-activity data, but they make recent antibiotic use an unlikely sole explanation for the main AGP IBD result (Supplementary Table S9). The selected IBD labels were also stable when compared against the pure-policy dCST baseline, indicating that the primary *Bacteroides*-centered findings are not artifacts of the absorb low-frequency policy (Supplementary Table S12).

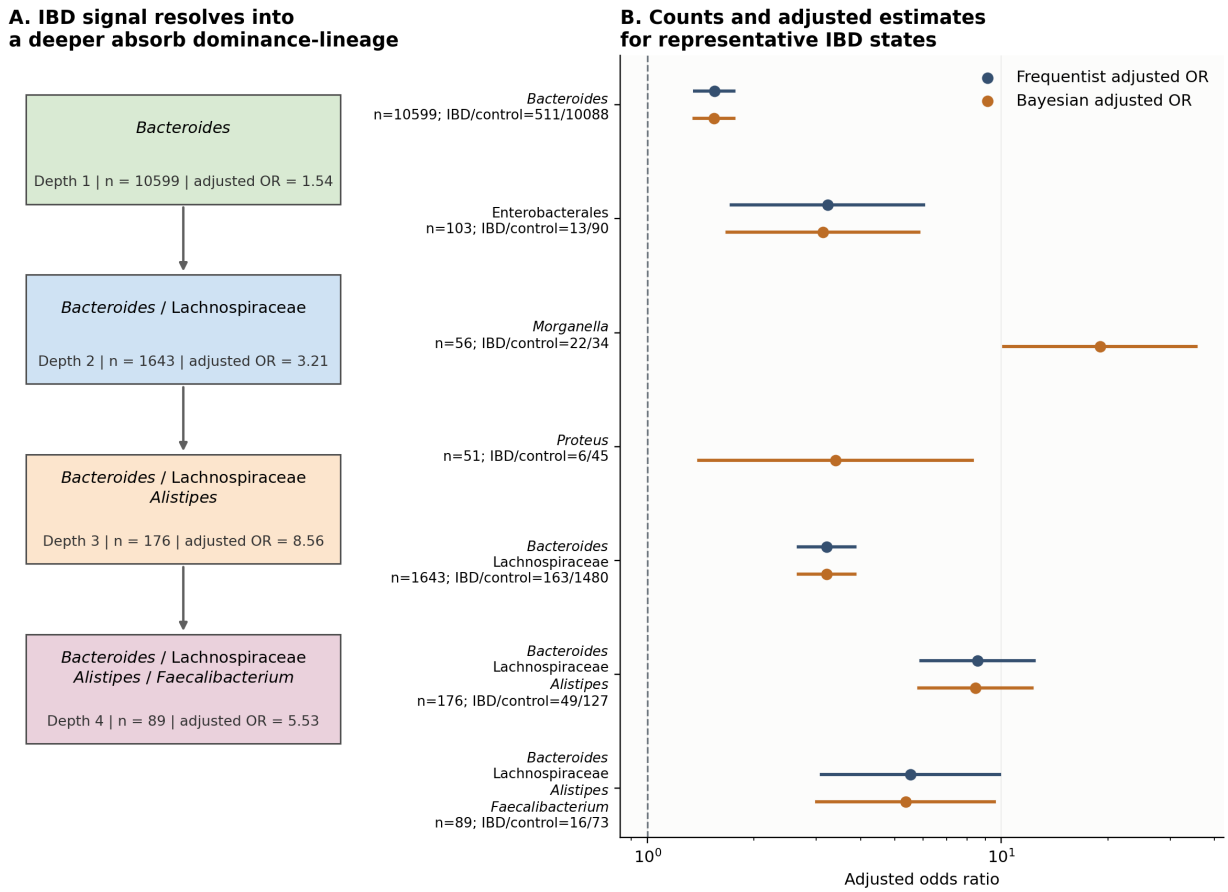

Figure 3: IBD-associated dominance-lineages resolve across the absorb dCST hierarchy. Panel A shows the progression from the broad depth-1 *Bacteroides* state to the depth-2 through depth-4 IBD-associated dominance-lineages emphasized in the main text. Panel B compares frequentist and Bayesian adjusted odds ratios for representative IBD-associated dominance-lineages and reports the corresponding IBD/control counts. The common dominance-lineages are supported by both the frequentist and Bayesian analyses. Rare enterobacterial dominance-lineages, including *Morganella* and *Proteus*, are included as sparse-label examples: Bayesian estimates remain finite, whereas the primary adjusted frequentist screen skips labels with insufficient support.

External validation used the two complementary modes defined in Methods: rebuilt-cohort analysis and direct transfer of AGP-derived absorb dCST labels.

Across the completed inflammatory cohorts, rebuilt-cohort validation was strongest in Halfvarson 2017, where the `IBD_vs_Healthy`, `Crohn_vs_Healthy`, and `UC_vs_Healthy` comparisons all remained significant after multiple-testing correction. The leading rebuilt labels differed across those contrasts but pointed to the same broad inflammatory axis: depletion of *Segatella* sp. in pooled IBD and Crohn disease together with enrichment of *Bacteroides dorei* in ulcerative colitis. PRJEB84421 provided narrower support through `OFG_vs_Healthy`, where a depth-2 *Faecalibacterium*/*Bacteroides* dominance-lineage was depleted ($q = 0.019$). By contrast, rebuilt `IBD_vs_Healthy` in HMP2 and rebuilt `Crohn_vs_Healthy` in Gevers 2014 remained directional but did not retain BH-significant absorb dCST dominance-lineages, which is not surprising given repeated sampling,

substantial clinical heterogeneity, and the more specialized ecology of those cohorts.

Direct transfer of AGP-derived absorb dCST labels was stricter but still informative. Halfvarson remained clearly positive when samples were mapped directly onto AGP-defined dominance-lineages: pooled IBD, Crohn, and UC comparisons all retained $q < 0.001$, with the transferred signal centered on enriched *Bacteroides* states and depleted *Faecalibacterium*-linked states. After explicitly coding pooled HMP2 inflammatory bowel disease as one case label, HMP2 also retained a weaker but significant depth-2 transferred *Bacteroides*/*Faecalibacterium* depletion in `IBD_vs_Healthy` ($q = 0.027$). Gevers 2014 and PRJEB84421 both mapped well into the AGP hierarchy but did not retain BH-significant transferred states, and the current Jacobs IBS cohort was likewise directional at most. We therefore interpret the present external data as strongest support for IBD-associated dominance-pattern reproducibility, with exact label portability clearly demonstrated in Halfvarson and supported more selectively in pooled HMP2 (Figure 4; Supplementary Table S4).

Because Halfvarson and HMP2 contain repeated samples, we also repeated the leading validation summaries after retaining one detected sample per subject. This sensitivity analysis narrowed the interpretation. The Halfvarson sample-level transfer results remained strongest overall, but the pooled IBD and Crohn transfer contrasts attenuated to suggestive rather than corrected-significant subject-level evidence, whereas the UC transfer signal remained significant ($q = 0.0029$). In HMP2, the transferred depth-2 *Bacteroides*/*Faecalibacterium* signal remained significant in the subject-level sensitivity analysis ($q = 0.012$). Direct transfer also showed phenotype-associated mapping differences in Halfvarson, especially at deeper depths, where IBD cases mapped more often than controls. We therefore treat direct AGP-derived label transfer as strongest when both the association and the mapping diagnostics are favorable, and we avoid interpreting nonmapping as biological absence (Supplementary Tables S10 and S11).

Table 1: External validation summary for the main IBD-centered dominance-state summaries. Rebuilt validation tests whether disease-linked dominance structure is rediscovered when the hierarchy is fit within an external cohort; direct AGP-derived label transfer tests whether the discovery-cohort labels themselves remain informative when assigned directly to external samples.

Cohort	Validation context	Evidence summary	Interpretation
Halfvarson 2017	Stool IBD, Crohn, and UC comparisons versus healthy controls; longitudinal sampling.	Sample-level rebuilt-cohort and AGP-derived label-transfer signals in all three inflammatory comparisons; subject-level sensitivity attenuated pooled IBD and Crohn transfer but retained UC transfer.	Strongest sample-level evidence, with subject-level checks supporting a calibrated label-portability claim.
HMP2 / IBDMDB	Pooled IBD versus healthy after explicit case recoding; mucosal gut-site samples with repeated longitudinal sampling.	Rebuilt mode was directional only, but AGP-derived label transfer retained a weaker depth-2 *Bacteroides*/ *Faecalibacterium* signal at sample level and after one-sample-per-subject sensitivity.	Selective transfer support, not a broad independent rediscovery.
PRJEB84421	Orofacial granulomatosis versus healthy controls.	Rebuilt depth-2 *Faecalibacterium*/ *Bacteroides* depletion retained significance ($q = 0.019$), but AGP-derived label transfer did not.	Narrow inflammatory support without label-level portability.
Gevers 2014	Pediatric stool Crohn disease versus healthy controls.	Samples mapped into the AGP hierarchy, but neither rebuilt-cohort nor AGP-derived label-transfer tests retained corrected absorb dCST dominance-lineage signal.	Informative mapping but not a positive replication.
Jacobs 2023 IBS 250bp	Baseline IBS versus healthy controls.	Directional at most, with no corrected AGP-derived label-transfer signal.	IBS validation remains inconclusive.

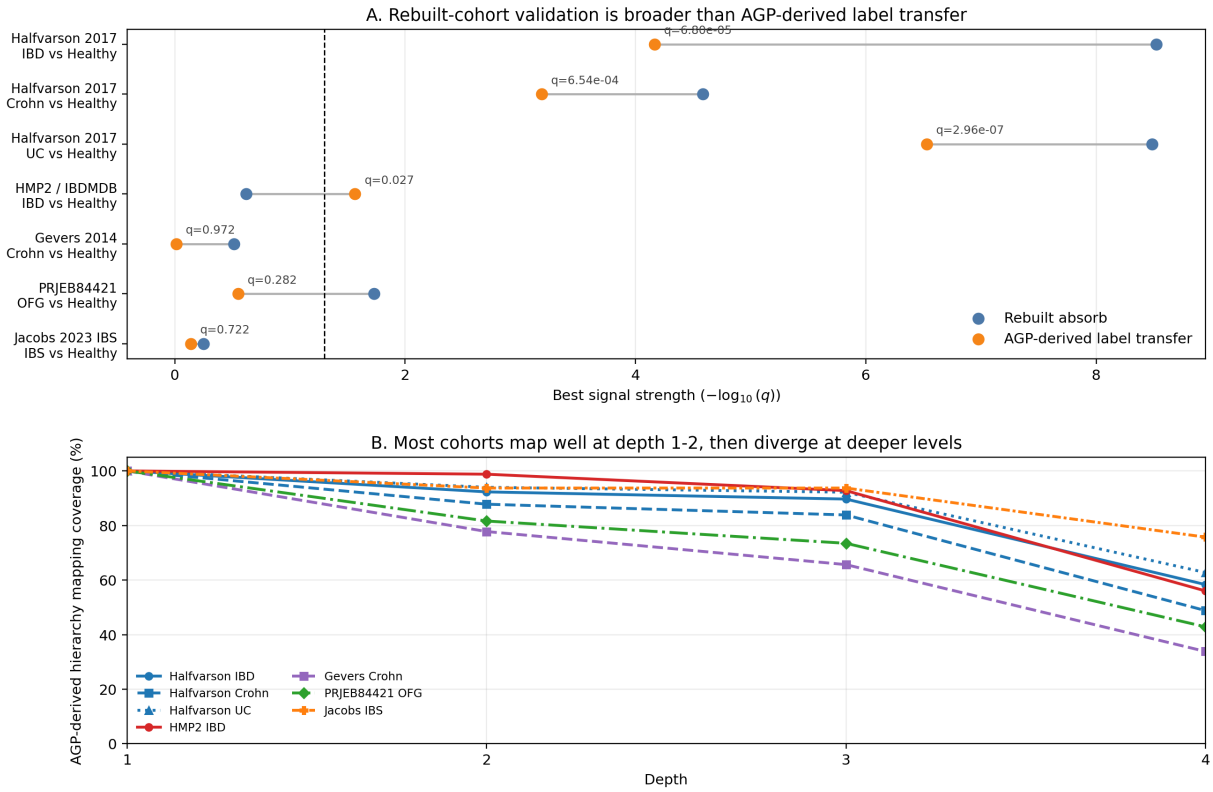

Figure 4: External validation separates reproducibility of the IBD-associated dominance pattern from exact label portability. Panel A compares the strongest BH-adjusted sample-level signal for each cohort-comparison under rebuilt-cohort absorb validation and direct AGP-derived absorb dCST label transfer. Panel B shows AGP-derived hierarchy mapping coverage across depths 1 to 4 for the same comparisons.

Irritable bowel syndrome (IBS)

IBS was included as a secondary, hypothesis-generating gut phenotype because its microbiome signal is typically subtler than in overt inflammatory bowel disease (Pittayanon et al. 2019; Dinan and Cryan 2015; Cryan and Dinan 2012). In the absorb dCST structure, IBS showed a modest but internally consistent shift toward enterobacterial dominance. At depth 1, *Escherichia-Shigella* was enriched in IBS (adjusted OR 1.24, q = 0.002), while *Staphylococcus*, *Faecalibacterium*, and *Prevotella_9* were depleted. At depth 2, two *Escherichia-Shigella*/*Bacteroides*-centered states remained significant, and by depth 3 the clearest surviving frequentist signal was an *Escherichia-Shigella*/*Bacteroides*/Lachnospiraceae dominance-lineage (adjusted OR 1.78, q = 0.018).

The Bayesian follow-up preserved the same ecological pattern at greater depth, with strong posterior support for depletion of *Faecalibacterium*- and *Prevotella_9*-linked states and for enrichment of deeper *Escherichia-Shigella*/*Bacteroides*/Lachnospiraceae dominance-lineages even when the frequentist q-values became sparse. These secondary results are consistent with a milder dysbiosis-associated dominance shift rather than evidence for a distinct IBS-specific ecological program, in line with the current IBS literature (Pittayanon et al. 2019; Tap et al. 2017).

Autoimmune diseases

The AGP autoimmune phenotype is an umbrella label rather than a single disease and therefore requires more cautious interpretation than IBD. Even so, autoimmune disease is an important category because the literature repeatedly links loss of immune-regulatory commensals and expansion of pathobionts to altered Treg/Th17 balance (Scher et al. 2013; Atarashi et al. 2013; De Luca and Shoenfeld 2019; Vatanen et al. 2018).

The broad AGP autoimmune phenotype provided a secondary, exploratory immune-associated comparison. Its absorb pattern was compatible with inflammation-associated dysbiosis but more heterogeneous than IBD. At depth 1, the dominant signals were enrichment of *Bacteroides* (adjusted OR 1.40, q = 1.6e-12) together with depletion of *Prevotella_9*, *Streptococcus*, *Faecalibacterium*, and *Staphylococcus*. At depth 2, the hierarchy sharpened around *Bacteroides*-centered dominance-lineages: *Bacteroides*/Lachnospiraceae reached adjusted OR 2.15 (q = 5.1e-22), *Bacteroides*/[*Eubacterium*] *eligens_group* reached adjusted OR 21.95 (q = 3.2e-20), and *Bacteroides*/*Alistipes* reached adjusted OR 1.70 (q = 2.5e-05). The same pattern continued deeper, with *Bacteroides*/Lachnospiraceae/*Alistipes* reaching adjusted OR 5.52 (q = 3.9e-21) at depth 3 and *Bacteroides*/Lachnospiraceae/*Alistipes*/*Faecalibacterium* remaining significant at depth 4 (adjusted OR 3.44, q = 6.1e-05).

The Bayesian analysis again helped mostly at the deeper levels. It supported the same major *Bacteroides*-centered dominance-lineages as the frequentist analysis, but it also retained support for additional depth-4 states such as *Bacteroides*/*Akkermansia*/Lachnospiraceae/*Faecalibacterium* and depleted *Bacteroides*/*Faecalibacterium*/Lachnospiraceae/*Agathobacter* states that no longer crossed the conventional q-value threshold. The pattern is directionally similar to IBD but requires more cautious interpretation: this umbrella phenotype appears enriched for an inflammatory *Bacteroides*-centered ecology, yet the pooled autoimmune label is too broad to support a disease-specific mechanistic claim. These findings are best interpreted as evidence for a shared inflammatory ecological axis rather than one causal autoimmune taxon (Scher et al. 2013; Vatanen et al. 2018; Atarashi et al. 2013; De Luca and Shoenfeld 2019).

Materials and Methods

The discovery analysis used AGP gut samples from the PRIME-derived SILVA gut-project abundance table (McDonald et al. 2018; Quast et al. 2013). We restricted the analysis to the 34,711 AGP rows present in the PRIME gut-project abundance table. Samples with total library size below 1,000 reads were excluded, removing 4,421 samples. Taxa were then retained only if they reached a count of at least 2 in at least 5% of the retained samples, corresponding to a prevalence threshold of 1,515 samples in this cohort. This feature filter retained 274 taxa, and no additional samples became all-zero after feature filtering, leaving 30,290 samples for the discovery analysis. Age was available for 27,372 samples (90.4%), sex for 25,598 (84.5%), and BMI for 26,541 (87.6%), producing 23,198 complete cases for the absorb-policy dCST dominance-lineage follow-up models. Because dCST construction was applied after this feature-retention step, all dCST labels should

be interpreted as retained-feature dominance labels. They do not imply dominance over every raw feature removed during preprocessing. Feature labels use the deepest informative SILVA annotation available for each retained feature, so labels can refer to species, genera, families, orders, or unresolved features such as *Unassigned*.

Phenotype flags were derived directly from the AGP `Phenotype` metadata field by fixed substring matching. The five outcome variables analyzed in the discovery screen were coded from the phrases `irritable bowel syndrome`, `inflammatory bowel disease`, `autoimmune`, `acid reflux`, and `seasonal allergies`. For each outcome, samples without the corresponding outcome-defining phrase served as controls, and outcome flags were therefore not mutually exclusive. In the filtered cohort, case counts were 3,105 for IBS, 1,166 for IBD, 2,872 for autoimmune disease, 2,859 for acid reflux, and 9,493 for seasonal allergies; corresponding complete-case counts after requiring age, sex, and BMI were 2,551, 993, 2,426, 2,279, and 8,284, respectively (Supplementary Table S1). For the AGP discovery cohort, IBD subtype, disease activity, medication exposure, and detailed treatment histories were not available in a form that supported clinical subtype modeling; the discovery IBD analyses are therefore interpreted as self-reported phenotype associations rather than as treatment-naive clinical subtype analyses. The coarser AGP `Antibiotic_Status` field was available for 22,939 of 23,198 age/sex/BMI complete-case samples and was used in sensitivity analyses that either added antibiotic status as a categorical covariate or excluded samples reporting antibiotic use in the previous week, month, or six months (Supplementary Table S9).

Absorb-policy dCSTs were constructed from the within-sample dominance order (Figure 1). A depth-1 state is defined by the dominant retained taxon, and deeper states are defined by recursively appending the second-ranked, third-ranked, and subsequent retained taxa. Because assignment depends only on within-sample rank order, the label is unchanged whether the input is expressed as raw counts, relative abundances, or any other sample-wide rescaling that preserves the order of taxa within each sample. We used the `absorb` low-frequency policy throughout. At depth 1, only dominance patterns observed in at least $n_0 = 50$ samples were kept as named states. Samples in smaller patterns were reassigned to the retained state with the largest abundance among the retained taxa for that sample, with deterministic tie handling. We then refined the hierarchy with $n_0 = 25, 12, 10, 8$, and 5 for depths 2 through 6, respectively. At each depth, the same rule was applied within the current parent state, producing ordered dCST dominance-lineages. Depths 1 through 4 are treated as the primary results in this manuscript, with deeper levels used mainly to judge whether the hierarchy remained structured.

The pure-policy dCST hierarchy without absorption served as a baseline comparison and is summarized in the supplement. The inferential workflow reported here is therefore absorb based from the cohort overview through phenotype-specific follow-up.

To ask whether a phenotype was associated with the overall absorb dCST dominance-lineage distribution at a given depth, we first applied Fisher's exact test to the phenotype-by-state table for that depth. We screened panels with 10,000 Monte Carlo draws and reran the strongest panels with 100,000 draws to obtain finer p-value resolution. We summarize the magnitude of these depth-level associations with Cramer's V. We then performed label-level follow-up in two parallel ways for

IBD, IBS, autoimmune disease, acid reflux, and seasonal allergies. The conventional analysis used Fisher tests and logistic regression adjusted for age, sex, and BMI. The Bayesian analysis used two complementary models. For unadjusted label-level contrasts we used a beta-binomial prevalence model with a Jeffreys prior, $\mathrm{Beta}(0.5, 0.5)$, summarized by the posterior median odds ratio, 95% credible interval, and posterior probability that the odds ratio exceeded 1. For adjusted analyses we fit Bayesian logistic regression models with `arm::bayesglm` using the same covariates as the frequentist regression and the package's default weakly informative prior. Those adjusted Bayesian fits were summarized by the exponentiated coefficient estimate, an approximate 95% credible interval based on the posterior standard error, and $\Pr(\beta > 0)$ for enrichment. Multiple testing in the conventional analyses was controlled with the Benjamini-Hochberg procedure (Benjamini and Hochberg 1995). We treat BH-adjusted frequentist results as the primary evidence for screen-level discovery and use Bayesian estimates as complementary stabilized summaries, especially for sparse labels where maximum likelihood estimates can be unstable or prespecified support filters exclude labels from the primary adjusted screen.

For the primary IBD analysis, we repeated the follow-up after excluding non-IBD controls with the other AGP phenotype strings analyzed in this study (IBS, autoimmune disease, acid reflux, or seasonal allergies). Selected IBD labels were also compared under the absorb and pure-policy hierarchies to check whether the main signals depended on absorption of low-frequency dominance patterns (Supplementary Table S12). External validation was evaluated in two complementary ways. In rebuilt-cohort validation, each external matrix was filtered and analyzed independently with the same absorb workflow used for the discovery cohort. In direct AGP-derived label-transfer validation, external samples were mapped onto the AGP-derived absorb dCST hierarchy after aligning each cohort's taxa to the AGP retained-feature dictionary. The completed validation cohorts included HMP2 / IBDMDB, Halfvarson 2017, Gevers 2014, PRJEB84421, and Jacobs 2023 IBS 250bp; the main text emphasizes the IBD-focused cohorts because they provided the clearest external readout. For cohorts with repeated subjects, we repeated leading validation tests after retaining one detected sample per subject, and for direct AGP-derived label transfer we summarized case/control mapping rates by depth (Supplementary Tables S10 and S11). The full comparison summary is given in Supplementary Table S4. All manuscript figures, tables, and analysis scripts were generated from workflows built around the `linf` R package, with project-specific runners maintained in the companion `gut_microbiome` repository.

Discussion

This study does not identify a new IBD dysbiosis axis. Instead, it represents established IBD-associated gut dysbiosis as dominance states that are interpretable at the level of individual samples. The central pattern is a broad *Bacteroides* background that resolves into deeper *Bacteroides*-centered dominance-lineages, together with sparse enterobacterial states that remain exploratory because of their small support and the absence of AGP disease-activity metadata. These labels should not be interpreted as evidence that dominant taxa alone explain IBD. Rather, they provide a compact way to name recurrent ecological contexts in which IBD cases are overrepresented.

The contribution of this application is therefore representational and analytical. The deterministic dCST label provides a rank-based sample-level state, the absorb hierarchy preserves named deeper dominance-lineages instead of sending a growing fraction of samples into a residual tail, and the Bayesian layer keeps sparse-label estimates finite while leaving BH-adjusted frequentist results as the primary discovery framework. The secondary IBS and autoimmune analyses provide related gut-outcome context rather than co-equal discovery claims; both were directionally compatible with published dysbiosis patterns but weaker or more heterogeneous than IBD.

Several limitations remain important. External validation supports IBD-focused dominance-pattern reproducibility more strongly than universal label portability: the strongest evidence comes from Halfvarson, HMP2 provides selective transferred support, and several cohorts mainly demonstrate mapping coverage rather than corrected label-level replication. Repeated-subject sensitivity and case/control mapping balance further argue against treating the current AGP-derived label dictionary as universally transferable. Broader stool-based validation will be needed, especially for IBS and any non-IBD extensions of this approach. AGP phenotypes are also self-reported and cross-sectional, so residual confounding, medication effects, antibiotic exposure, and reverse causation remain plausible even though the leading IBD signals were robust to the coarse antibiotic-status checks available here. The dCST labels are also retained-feature labels constructed after prevalence filtering, and some low-resolution or potential non-gut labels appear in the full hierarchy; those states are retained for transparent accounting but require caution as biological signals. These concerns are especially important for supplementary non-IBD phenotypes. The current evidence nevertheless supports absorb-policy dCSTs as a biologically interpretable analysis layer and motivates broader testing of their generalizability in independent cohorts.

Conclusion

Taken together, these results support dCSTs as a compact, interpretable way to summarize established IBD-associated gut dysbiosis as sample-level dominance structure. The current evidence is strongest for IBD, more selective for cross-cohort label portability, and exploratory for broader gut-outcome applications.

Data and Code Availability

The manuscript source and paper assets are maintained in the public `linf` repository: https://github.com/pgajer/linf. The project-specific discovery and follow-up runners are maintained in the public `gut_microbiome` repository: https://github.com/pgajer/gut_microbiome. The discovery analysis used the American Gut Project stool cohort (`PRJEB11419`) through PRIME-derived gut abundance tables. The main project scripts are:

- `run_agp_absorb_depthscan.R`, which generated the absorb dCST hierarchy;
- `run_absorb_label_followup.R`, which generated the phenotype-specific frequentist and Bayesian follow-up analyses for absorb-policy dCST dominance-lineages;

- `run_external_case_control_validation_absorb.R`, which generated the rebuilt-cohort validation results;
- `run_external_case_control_validation_frozen_agp.R`, which generated the direct AGP-derived label-transfer validation results;
- `export_prime_project_validation_inputs.py`, which generated PRIME-derived HMP2 input exports.
- `build_ibd_dominance_lineage_profile_table.py`, which generated the retained-feature abundance profile table for the main IBD dCST dominance-lineages.
- `build_ibd_reviewer_sensitivity_assets.R`, which generated the antibiotic-status, subject-level validation, mapping-balance, absorb-policy robustness, and taxonomic-rank sensitivity tables added for reviewer calibration.

Supplementary materials summarize phenotype definitions and sample counts, the pure-versus-absorb depth-change summary, IBD sensitivity analyses, retained-feature dominance-lineage profiles, selected reference analyses retained outside the main text, and the cross-cohort validation and mapping-balance comparisons.

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Competing Interests

The authors declare no competing interests.

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